Ch 6.1: Subset Selection

Lecture 17 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Fri, Oct 13, 2023

Announcements

Last time

- Boostrapping

Covered in this lecture

- Subset selection
- Forward and Backward Selection

12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4 Due
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	Hw #5
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	

Announcements:

- HW #5 posted and due Wednesday October 18

Section 1

Last time

Goals of fitting a given model

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation. → minimize MSE $= \frac{1}{n}$ RSS

Prediction accuracy

- How good a job am I doing?
- In the case of linear stuff:

  If the true relationship is linear
  ⇒ least squares estimates have low bias

  If in addition $n \gg p$ ⇒ least squares has low variance

- If n is not much larger than p
  ↳ high variability
  overfitting
  poor predictions
- If $n < p$
  no least squares solution

Goal: Shrink / Constrain # of input variables (p)

Goals of fitting a given model

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation.

Model Interpretability

- Often, many variables not related to response
- Including them leads to unnecessary complexity
- Idea: Don't use them!
  - ↳ or set coeff to 0

Goal: Use fewer inputs ($\downarrow p$), use the ones that help

Goal of next chapter

- Subset selection (today)
  - ↳ Pick a good subset!
- Shrinkage methods (M+W)
  - ↳ force some coeff to be 0
- Dimensionality reduction (After the test)
  - ↳ linear combos of inputs to have fewer inputs

Section 2

Best Subset Selection

Too many variables

All subsets of 4 variables ($2^4 = 16$)

- $\emptyset$ M_0

$k=1$

- X_1
- X_2
- X_3 $= M_1$
- X_4

M_1

$k=2$

- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$ M_2
- $X_2\ X_3$
- $X_2\ X_4$
- $X_3\ X_4$

$k=3$

- $X_1\ X_2\ X_3$
- $X_1\ X_2\ X_4$
- $X_1\ X_3\ X_4$
- $X_2\ X_3\ X_4$ M_3

$k=4$

- $X_1\ X_2\ X_3\ X_4$ M_4

Idea: Run the model using every subset of input variables
Return the best one.

One way of breaking this up

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$: # of input variables
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, ~~or equivalently largest R^2~~.
   MSE
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, ~~C_p (AIC), BIC, or adjusted R^2.~~

Group work: calculate by hand

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the R^2 value computed for the model learned using each possible subset of variables.

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

M_0 → Null model

M_1 → X2

M_2 → X3,X4

M_3 → X1,X3,X4

M_4 → X1,X2,X3,X4

M_2 is the winner!

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using best subset selection?
2. What subset of variables is returned using best subset selection?

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

- $\emptyset$

- X_1
- X_2
- X_3
- X_4

- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$
- $X_2\ X_3$
- $X_2\ X_4$
- $X_3\ X_4$

- $X_1\ X_2\ X_3$
- $X_1\ X_2\ X_4$
- $X_1\ X_3\ X_4$
- $X_2\ X_3\ X_4$

- $X_1\ X_2\ X_3\ X_4$

Section 3

Forward Selection

What's the problem?

- Checking 2^p model
  - ↳ Training
  - ↳ not reasonable for large p $(p > 40)$

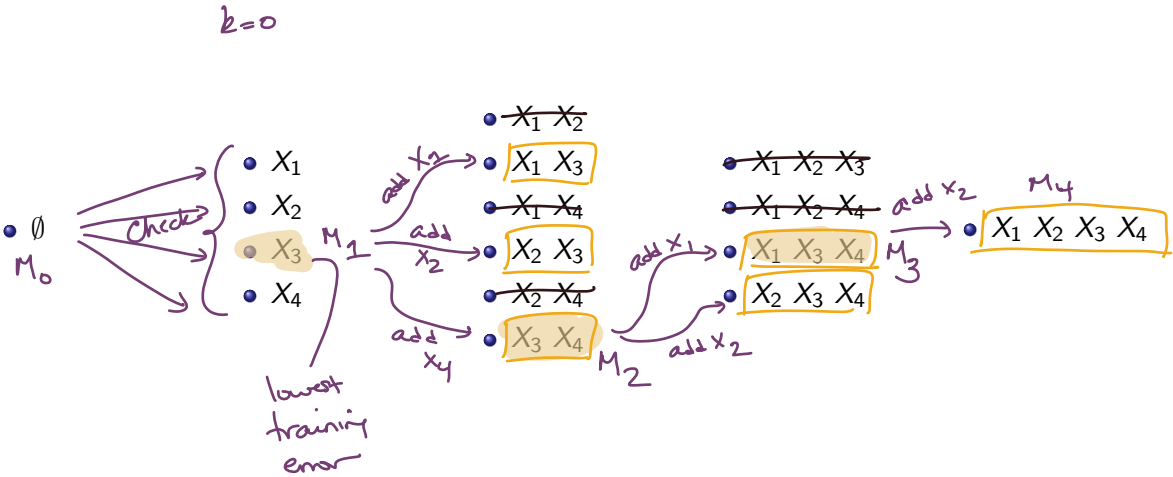
Forward Stepwise Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS ~~or highest R^2~~.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, ~~C_p, AIC, BIC, or adjusted R^2~~.

Now have $M_0, \ldots, M_p$

An example for Forward Stepwise Selection



Group work: by hand same example with forward example

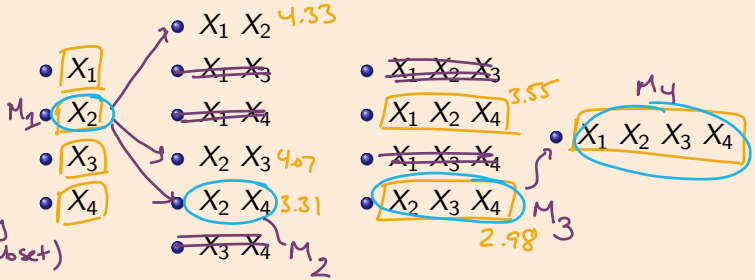
We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the R^2 value computed for the model learned using each possible subset of variables.

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using forward selection?
2. What subset of variables is returned using forward subset selection?

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39



X_2 X_3 X_4 is returned as the best subset

Pros and Cons of Forward Stepwise

Pros:

Computationally cheaper than all subsets

# of models fit:

$$\underbrace{1}_{\text{\# fit for } M_0} + \sum_{k=0}^{p-1} \underbrace{(p-k)}_{\text{\# fit for find } M_k} = 1 + \frac{p(p+1)}{2} \ll 2^p$$

Way better than 2^p

Cons:

- Not guaranteed to find the best model
- Our example found different subset

Section 4

Backward Selection

Backward stepwise selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors. drop one
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS ~~or highest R^2~~.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, ~~C_p (AIC), BIC, or adjusted R^2~~.

An example for Backward Stepwise Selection

- $\emptyset$

- X_1
- X_2
- X_3
- X_4

- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$
- $X_2\ X_3$
- $X_2\ X_4$
- $X_3\ X_4$

- $X_1\ X_2\ X_3$
- $X_1\ X_2\ X_4$
- $X_1\ X_3\ X_4$
- $X_2\ X_3\ X_4$

- $X_1\ X_2\ X_3\ X_4$

Group work: by hand same example with backward

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the R^2 value computed for the model learned using each possible subset of variables.

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using forward selection?
2. What subset of variables is returned using forward subset selection?

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

M_0

$\emptyset$

M_1

$\cancel{X_1}$

$\cancel{X_2}$

X_3 8.16

X_4 8.33

$\cancel{X_1}\ \cancel{X_2}$

$X_1\ X_3$ 5.82

$X_1\ X_4$ 3.17

$\cancel{X_2}\ \cancel{X_3}$

$\cancel{X_2}\ \cancel{X_4}$

$X_3\ X_4$

M_2 3.06

$X_1\ X_2\ X_3$

$X_1\ X_2\ X_4$

$X_1\ X_3\ X_4$

$X_2\ X_3\ X_4$

M_3

$X_1\ X_2\ X_3\ X_4$

M_4

X_3, X_4

retained

Pros and Cons of Backward Stepwise

Pros:

Computationally cheaper

models fit is the same as forward Selection

Cons:

Not guaranteed to find "best"

Not possible at all if $n < p$

↳ in this case use forward to build up from small p

Code

TL;DR

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

- Modify step 2 with forward or backward selection
- Choose best model in step 3 using one of our adjusted training scores or CV

Next time

12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4 Due
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	HW#5
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	